

G001-SIM Series Specifications

G001-SIM-01 through SIM-08 Configurations

The G001-SIM series encompasses eight distinct simulator configurations, each designed to meet specific training requirements and certification levels. All variants utilize the common G001 airframe architecture but differ in motion systems, visual displays, and avionics fidelity.

Configuration Overview:

- **G001-SIM-01:** Level D full flight simulator with six-axis electric motion system, 200° x 40° cylindrical visual display at 4K resolution per channel, and complete systems replication including hydraulic loading on flight controls
- **G001-SIM-02:** Level D variant with hydraulic motion platform, identical visual specifications, enhanced for upset recovery training with expanded motion envelope
- **G001-SIM-03:** Level C fixed-base trainer with 180° x 40° visual system, functional cockpit procedures trainer without motion cueing
- **G001-SIM-04:** Level B flight training device with 150° forward visual, simplified systems modeling, suitable for instrument procedures and basic maneuvers
- **G001-SIM-05:** Part-task trainer focused on FMS/FMC operations, includes functional CDU, MFD, and PFD displays without full cockpit shell
- **G001-SIM-06:** Maintenance training simulator featuring complete systems panels, circuit breaker arrays, and BIT access for troubleshooting procedures
- **G001-SIM-07:** Cabin crew trainer with passenger cabin mockup, door trainer integration, and emergency systems activation
- **G001-SIM-08:** Integrated procedures trainer (IPT) combining cockpit and cabin environments for coordinated crew resource management scenarios

All configurations from SIM-01 through SIM-04 include the standard G001 avionics suite with dual FMS, EICAS/ECAM displays, and IRS alignment capabilities. Motion-equipped variants (SIM-01 and SIM-02) feature transport delays not exceeding 150 milliseconds and acceleration onset cueing within regulatory tolerances. Visual systems across all configurations maintain minimum 60Hz refresh rates with collimated infinity focus where applicable. Each simulator connects to the central instructor operating station via redundant fiber-optic network architecture supporting real-time performance monitoring and fault injection capabilities.

Simulator Type Classifications (Full Flight, Fixed Training, CPT)

The G001-SIM Series encompasses multiple certification levels designed to meet varying training requirements and regulatory compliance standards. Each classification adheres to specific technical and operational criteria as defined by ICAO Document 9625 and applicable national aviation authority regulations including FAA 14 CFR Part 60 and EASA CS-FSTD(A).

Full Flight Simulator (FFS) - Level D Configuration

The G001-SIM FFS represents the highest fidelity classification, incorporating a complete six-degree-of-freedom motion system with electric actuators providing ± 24 inches of heave, ± 18 inches of lateral/longitudinal displacement, and full angular motion envelope. The visual system delivers 200° horizontal by 40° vertical field-of-view with minimum 6-channel projection at 1920x1200 resolution per channel. All aircraft systems are modeled to full operational depth, including hydraulic pressure transients, bleed air distribution dynamics, and electrical load management with realistic failure propagation. Sound cueing system provides 32-channel spatial audio with subwoofer integration for engine and airframe vibration replication.

Fixed Training Device (FTD) - Level 5/6 Configuration

The G001-SIM FTD eliminates the motion platform while maintaining high-fidelity aerodynamic modeling and systems simulation. Visual display utilizes 180° horizontal FOV with three-channel projection. Flight control loading system replicates force feedback characteristics within $\pm 5\%$ of aircraft data. All primary flight instruments, navigation systems (FMS, IRS, GPS), and communication equipment function identically to FFS configuration. Suitable for procedural training, instrument approaches, and normal/abnormal checklist procedures.

Cockpit Procedures Trainer (CPT)

The CPT configuration focuses on systems familiarization and cockpit flow training without aerodynamic modeling or visual system requirements. Features include:

- Functional overhead, glareshield, and pedestal panels
- Operational FMC/CDU with navigation database
- EICAS/ECAM display simulation
- Electrical, hydraulic, and pneumatic system logic
- APU start/shutdown sequences

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- IRS alignment procedures (real-time and fast modes)

All configurations support instructor operating station (IOS) control with malfunction insertion capability, weather injection, and automated performance monitoring systems.

Site Locations and Lead Engineer Contacts

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The G001-SIM Series full-flight simulators are currently deployed across seven authorized training facilities worldwide. Each site maintains dedicated technical support staff with direct escalation paths to the OEM engineering team. Site-specific configuration data, including motion system calibration parameters, visual system alignment matrices, and instructor operating station (IOS) software versions, are maintained in the centralized Configuration Management Database (CMDB) accessible via the secure portal at simtech.g001aviation.com.

Primary Installation Sites

North American Operations Center - Phoenix, Arizona, USA - Lead Engineer: Michael Chen, P.Eng - Contact: +1-602-555-0147 | m.chen@g001sim.com - Simulators: G001-SIM-003, G001-SIM-007, G001-SIM-012 - Level D qualification maintained under FAA NSPM authorization

European Training Hub - Frankfurt, Germany - Lead Engineer: Dr. Petra Hoffmann - Contact: +49-69-555-0283 | p.hoffmann@g001sim.eu - Simulators: G001-SIM-005, G001-SIM-009 - EASA FSTD(A) qualification current through Q2 2025

Asia-Pacific Center - Singapore - Lead Engineer: Raymond Tan - Contact: +65-6555-0891 | r.tan@g001sim.asia - Simulators: G001-SIM-011, G001-SIM-014 - CAAS Type VII certification with extended envelope authorization

Additional satellite facilities operate in Dubai (UAE), São Paulo (Brazil), Tokyo (Japan), and Melbourne (Australia) with regional support engineers reporting to the primary site leads.

Technical Support Escalation Procedure

1. Initial troubleshooting performed by on-site maintenance technician
2. Fault codes logged in IOS maintenance page (MAINT > FAULT LOG)
3. BIT results documented with timestamp and simulator state
4. Contact appropriate Lead Engineer if fault persists beyond Level 1 troubleshooting
5. Critical system failures (motion, visual, or flight model anomalies) require immediate notification to OEM duty engineer: +1-800-555-G001

All technical communications must reference simulator serial number, software load version, and active QTG test results where applicable.

Device Status Lifecycle (Active, Inactive, In Service)

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The G001-SIM Series implements a three-state device status model to manage simulator subsystem availability and operational readiness. Each Line Replaceable Unit (LRU) and software module maintains an independent status flag accessible through the Instructor Operating Station (IOS) and the Maintenance Control Panel (MCP). Understanding these states is critical for proper fault isolation and system configuration management.

Active Status indicates a device is powered, has completed its Built-In Test (BIT) sequence successfully, and is available for normal operations. Active devices respond to bus commands, generate appropriate ARINC 429 or ARINC 664 traffic, and participate in system-wide health monitoring. For example, an Active Inertial Reference System (IRS) will output present position, track, groundspeed, and attitude data at the specified refresh rate of 50Hz.

Inactive Status designates a device that is installed and powered but deliberately excluded from the simulation environment. This state is commonly used during partial-panel training scenarios or when simulating specific equipment failures. An Inactive Flight Management Computer (FMC) will not respond to CDU inputs, will not compute navigation solutions, and will trigger appropriate crew alerting system messages on EICAS or ECAM displays. The device remains in standby mode, consuming minimal power and maintaining configuration data.

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In Service Status represents a transitional state during which a device is undergoing initialization, alignment, or reconfiguration. This status applies during IRS alignment procedures, FMS database updates, or when recovering from a transient fault condition. Devices in this state may generate partial data outputs or exhibit degraded performance characteristics.

Status Transition Procedures

1. Access the MCP System Configuration page
2. Select the target LRU from the device tree
3. Verify current status indicator (green=Active, amber=In Service, white=Inactive)
4. Press the STATUS CHANGE softkey
5. Confirm the transition request via the EXECUTE prompt
6. Monitor the transition completion time (typically 5-30 seconds depending on device type)
7. Verify new status indication and check for any fault annunciations

Status changes are logged automatically in the maintenance event recorder with timestamps accurate to 10 milliseconds.